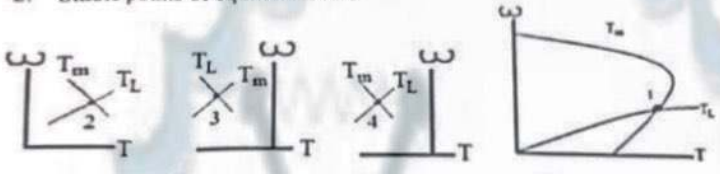
Q1: (20 marks)

Answer the following questions by circling the right answer in the table at the end of this question

1. Adding resistance in series with armature of dc motor will cause:

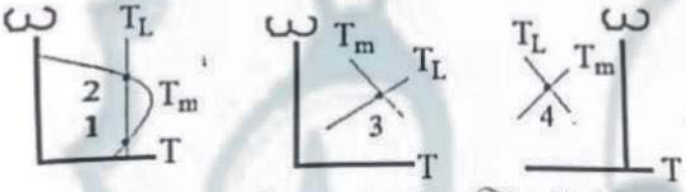
a. Increasing of motor torque
b. decreasing in efficiency
c. Increasing of short- circuit current
d. Increasing of starting torque

2. Stable points of equilibrium are



a. 1only b. 2 only c. 1 and 3 d. 1, 2 and 4

3. Stable points of equilibrium are



a. 1 and 2 b. 1 and 4 c. 2 and 3 d. 1 and 3

4. If the torque developed by the motor is $T = -T_L - J\frac{d\omega}{dt}$, the motor is

a. decelerated b. accelerated c. working at S.S.P d. none of these

5. In the electric drive system, the main function of the power converter is

a. matching the motor and MTD to meet the load requirements
b. adapting to the changes of the environment only
c. sensing the feedback variables
d. providing the motor with the suitable voltage signal

6. If the load is driven by two motors, the drive is

a. interconnected b. group
c. regulated d. auxiliary

7. The synchronous motor reaches the highest torque at

a. θ=20° b. θ=90° c. θ=30° d. θ=20°- 30°

8. The drive system input power is:

a. Thermal power b. Mechanical power
c. Electrical power d. Electromagnetic power

9. Synchronization conditions for the SM are

a. $\omega_{syn} \geq 0.9\omega_o, T_{syn} > T_L$
b. $\omega_{syn} \geq 0.9\omega_o, T_{syn} > T_n$
c. $\omega_{syn} \geq 0.95\omega_o, T_{syn} > T_n$
d. $\omega \geq \omega_{syn}, T_{syn} > T_L$

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10. The DC motor which has the best linear characteristics
a. separately excited b. compound c. shunt d. universal

11. The reactive load torque industrial machinery is:
a. crane b. metal -cutting c. hoist d. elevator

12. Production machineries load are
a. constant torque b. active c. constant power d. none of these

13. For D.C. series motor which of the following is correct
a. has soft characteristics b. field current is independent of load
c. can be started at no-load d. suitable for heavy duty starting

14. When varying load torque, separately excited dc motor has variable
a. speed b. armature current c. back emf d. all of these

15. Electric motor has moment of inertia of 1kg.m2, connected through gearbox of 5 gear ratio and 1kg.m2 to the drum with moment of inertia 4 kg.m2. The drive system total (equivalent) moment of inertia equals:
a. 6 kg.m² b. 2 kg.m² c. 2.16 kg.m² d. none of these

16. Crane motor lowers load of 10 N.m through gearbox, which has gear ratio of 10. If the system efficiency equals 0.9, the referred load torque to motor shaft is:
a. 1.11N.m b. 0.9N.m c. 11.1N.m d. 100 N.m

17. If the motor works in the 3dh quadrant. This mode of operation is a
a. reverse motoring mode b. forward regenerative mode
c. forward motoring mode d. reverse braking mode

18. Separately excited dc motor with the following ratings:
$P_{2N}=10kW, V_a=V_f=220V, \omega_N=100rad/s, R_a=0.4\Omega, I_{an}=50A, k\Phi_N=2V.s$ The motor efficiency is:
a. 0.9 b. 0.91 c. 0.88 d. 0.85

19. Separately excited dc motor with the following ratings:
$P_{2N}=10kW, V_a=V_f=220V, \omega_N=100rad/s, R_a=0.4\Omega, I_{an}=50A, k\Phi_N=2V.s$ The starting torque is:
a. 100 Nm b. 98Nm c. 1100 Nm 550Nm

20. Separately excited dc motor with the following ratings:
$V_a=V_f=200V, \omega_N=100rad/s, R_a=0.4\Omega, I_{an}=50A$ no-load speed is:
a. 105 rad/s b. 150 rad/s c. 111 rad/s 120 rad/s

Ques	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans	D	D	C	A	A	C	B	C	C	A	B	D	A	D	C	A	A	A	A	C
	A	V	V	A	A	V	V	A	V	V	A	A	A	V	A	V	A	A	V	
	B	D	C	B	D	A	B	C	D	A	B	C	D	A	C	B	A	B	C	

Q2 : Answer with yes or no (5 marks)

1. Series dc motors are self – excited (yes)
2. Brushless permanent magnet synchronous motors have distributed windings (no)
3. AC motors should be checked by maximum to nominal torque ratio (yes)
4. Synchronous motors can be working in the first quadrant only (no)
5. Squirrel-cage motor torque is proportional to the rotor current over the whole motoring mode (yes)

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Q3 : (5 marks)

A drive has following parameter

$J = 10kg.m^2, T = 15 + 0.05n\ (N.m), Tl = 5 + 0.06n\ (N.m)$, where n is the speed in rpm.

Initially the drive is working in steady state. Now the drive is braked by electrical braking. Torque of the motor in braking is given by T= -10-0.04n (N.m) . Calculate time taken by the drive to fully stop.

Solution :

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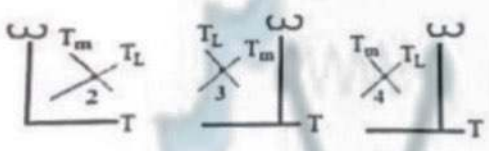
Q1: (20 marks)

Answer the following questions by circling the right answer in the table at the end of this question

1. Adding resistance in series with armature of dc motor will cause:
a. Increasing of motor torque (b) decreasing in efficiency
c. Increasing of short- circuit current d. Increasing of starting torque

1-B

2. Stable points of equilibrium are

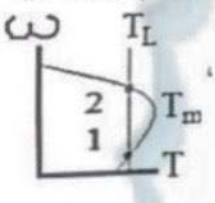


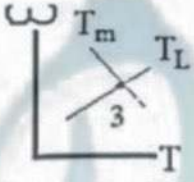


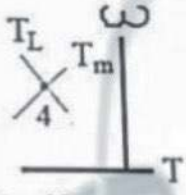
2-D

a. 1only b. 2 only c. 1 and 3 (d.) 1, 2 and 4

3. Stable points of equilibrium are







3-C

a. 1 and 2 b. 1 and 4 c. 2 and 3 (d.) 1 and 3

4. If the torque developed by the motor is $T = -T_L - J\frac{d\omega}{dt}$, the motor is
a. decelerated (b.) accelerated (c.) working at S.S.P d. none of these

4-B

5. In the electric drive system, the main function of the power converter is
a. matching the motor and MTD to meet the load requirements
b. adapting to the changes of the environment only
c. sensing the feedback variables
(d.) providing the motor with the suitable voltage signal

5-D

6. If the load is driven by two motors, the drive is
a. interconnected b. group
c. regulated (d.) auxiliary

6-A

7. The synchronous motor reaches the highest torque at
a. θ=20° b. θ=90° c. θ=30° d. θ=20°- 30°

7-B

8. The drive system input power is:
a. Thermal power b. Mechanical power
(c.) Electrical power d. Electromagnetic power

8-C

9. Synchronization conditions for the SM are
a. $\omega_{syn} \geq 0.9\omega_o, T_{syn} > T_L$ b. $\omega_{syn} \geq 0.9\omega_o, T_{syn} > T_n$
(c.) $\omega_{syn} \geq 0.95\omega_o, T_{syn} > T_n$ d. $\omega \geq \omega_{syn}, T_{syn} > T_L$

9-D

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10-A

10. The DC motor which has the best linear characteristics
a. separately excited b. compound c. shunt d. universal

11-B

11. The reactive load torque industrial machinery is:
a. crane b. metal -cutting c. hoist d. elevator

12-c

12. Production machineries load are:
a. constant torque b. active c. constant power d. none of these

13-D

13. For D.C. series motor which of the following is correct
a. has soft characteristics b. field current is independent of load
c. can be started at no-load d. suitable for heavy duty starting

14-A

14. When varying load torque, separately excited dc motor has variable
a. speed b. armature current c. back emf d. all of these

15-C

15. Electric motor has moment of inertia of 1kg.m2, connected through gearbox of 5 gear ratio and 1kg.m2 to the drum with moment of inertia 4 kg.m2. The drive system total (equivalent) moment of inertia equals:
a. 6 kg.m² b. 2 kg.m² c. 2.16 kg.m² d. none of these

16-B

16. Crane motor lowers load of 10 N.m through gearbox, which has gear ratio of 10. If the system efficiency equals 0.9, the referred load torque to motor shaft is:
a. 1.11N.m b. 0.9N.m c. 11.1N.m d. 100 N.m

17-A

17. If the motor works in the 3dh quadrant. This mode of operation is a
a. reverse motoring mode b. forward regenerative mode
c. forward motoring mode d. reverse braking mode

18-B

18. Separately excited dc motor with the following ratings:
$P_{2N}=10kW, V_a=V_f=220V, \omega_N=100rad/s, R_a=0.4\Omega, I_{an}=50A, k\Phi_N=2V.s$ The motor efficiency is:
a. 0.9 b. 0.91 c. 0.88 d. 0.85

19-C

19. Separately excited dc motor with the following ratings:
$P_{2N}=10kW, V_a=V_f=220V, \omega_N=100rad/s, R_a=0.4\Omega, I_{an}=50A, k\Phi_N=2V.s$ The starting torque is:
a. 100 Nm b. 98Nm c. 1100 Nm 550Nm

20. Separately excited dc motor with the following ratings:
$V_a=V_f=200V, \omega_N=100rad/s, R_a=0.4\Omega, I_{an}=50A$ no-load speed is:
a. 105 rad/s b. 150 rad/s c. 111 rad/s 120 rad/s

Ques	1 D	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans	B	D	C	A	A	C	B	C	C	A	B	D	A	D	C	A	A	A	A	C

B D C B D A B C D A B C D A C B A B C

Q2 : Answer with yes or no (5 marks)

1. Series dc motors are self – excited (yes)
2. Brushless permanent magnet synchronous motors have distributed windings (no)
3. AC motors should be checked by maximum to nominal torque ratio ( )
4. Synchronous motors can be working in the first quadrant only (yes)
5. Squirrel-cage motor torque is proportional to the rotor current over the whole motoring mode (yes)

1-T 2-T 3-T 4-F 5-F

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$J = 10\ kg \cdot m^2$, $T = 15 + 0.05n\ (N.m)$, $T_L = 5 + 0.06n\ (N.m)$

$T_{braking} = -10 - 0.04n\ (N.m)$

Solve :-

$n = 9.55w$

$T = 15 + 0.05(9.55w) \Rightarrow \boxed{T = 15 + 0.4775w}$

$T_L = 5 + 0.06(9.55w) \Rightarrow \boxed{T_L = 5 + 0.573w}$

$T_{braking} = -10 - 0.04(9.55w) \Rightarrow \boxed{T_{braking} = -10 - 0.382w}$

$T = T_L \rightarrow 15 + 0.4775w = 5 + 0.573w$

$w_1 = 104.7$

$T_{braking} - T_L = J \frac{dw}{dt}$

$dt = J \frac{dw}{T_b - T_L} \qquad t = \int \frac{J}{T_b - T_L} dw$

$t = \int_0^{104.7} \frac{10}{-15 - 0.955w} dw \qquad t = \boxed{12.37\ s}$

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